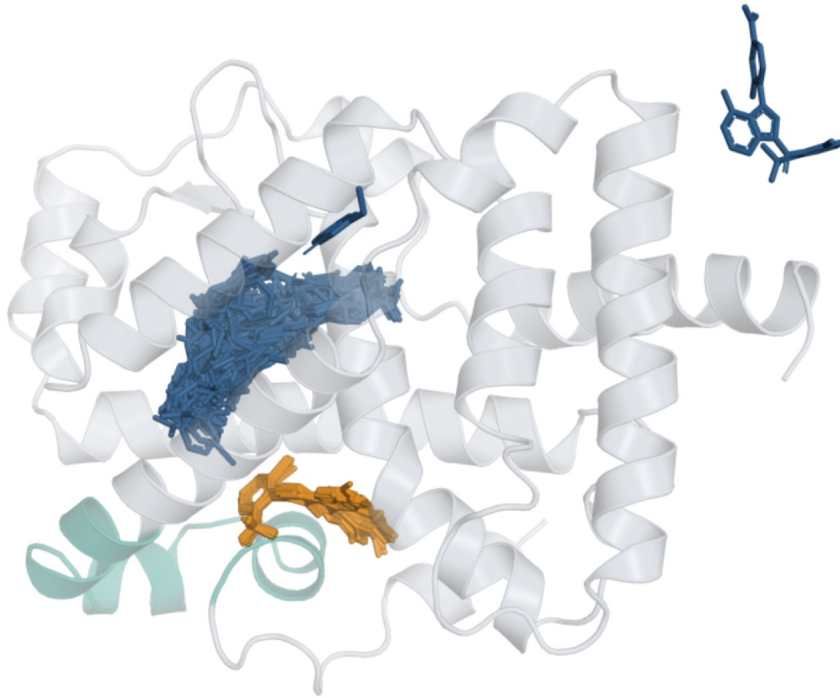


APHERIS FOUNDRY HACKATHON · TEAM-2

Ror against the machines

Experiments E0–E31 · one RTX PRO 6000 · 2026-10-02

Ligands bind in two places, and one of them is H12's seat



12.9 Å

between the two cluster centres — the sites are cleanly separated

0.27 Å

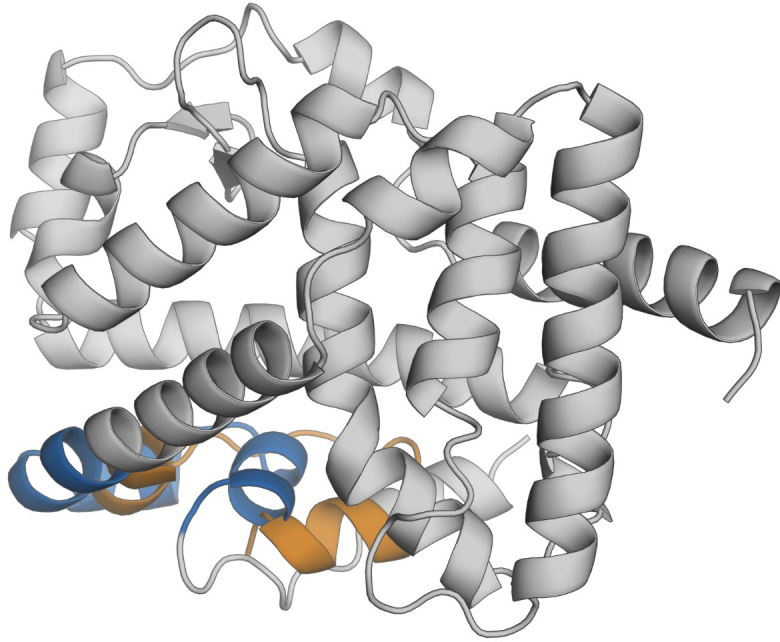
mean spread of the 21 allosteric ligands: one tight, specific pocket

The orange cluster and the teal helix occupy the same volume. A ligand there and a packed-in H12 cannot both exist.

A correlation in the deposited set, not a lever: forcing a ligand into either site does not move H12. We tested that four times.

All 89 ligands of the 78 deposited structures, superposed on one core. Blue: 68 orthosteric. Orange: 21 allosteric. Teal: H12 in the agonist position.

The label is a helix position



RORyt is always on. A drug has to switch it *off*, and that is a conformational event.

Helix 12 packed in → agonist

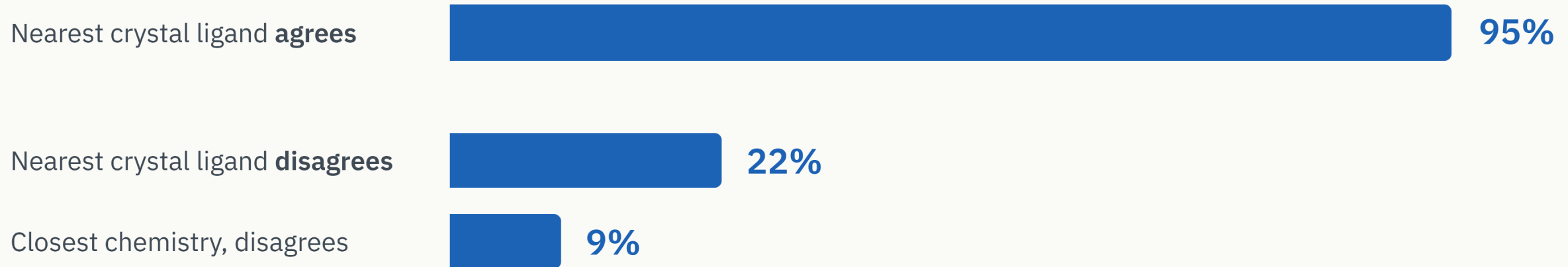
Helix 12 displaced → antagonist

9.7 Å apart

We measure it by superposing the core and reading H12 without refitting. The conventional crop metric scores a 10 Å displacement as 0.000 Å.

The model copies its nearest neighbor

150 labeled compounds, no deposited structure. Sorted by one question: does the crystal ligand each most resembles carry the same label, or the opposite?

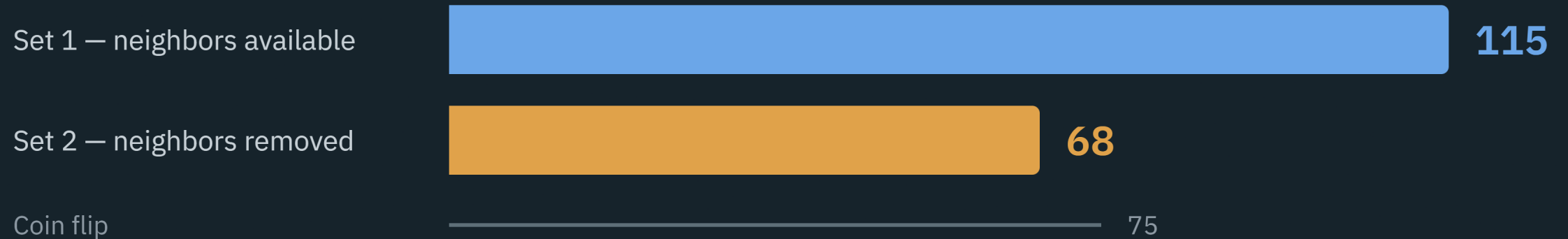


The neighbor predicts the model's answer better than the compound does — 136 matches against 115.

Share of each group whose predicted state matches the compound's own assay label · E5, n = 150

Remove the neighbors and the call collapses

A second set of 150, built to close the retrieval route: no shared scaffold and nothing within Tanimoto 0.5 of set 1, balanced 75/75 by label.



45 % — below chance. The structure model's own call cannot be the router on a blind target.

Correct H12 state calls of 150 · set 2 removes the *structure model's* neighborhood; the router's is untouched at median Tanimoto 0.716

Two levers, and they are switches

Coactivator peptide



75 / 75

agonist calls. It only docks when H12 is packed in.

6T4I template



75 / 75

antagonist calls. We spent days calling this a bug.

Measured on set 2 — the chemistry with no close neighbor. Generation is not the problem.

Every remaining point is in one decision

The whole pipeline scored against the 78 deposited structures, split by whether the call was right.

Call right — 49 targets

state correct 49 / 49

Call wrong — 28

state correct 0 / 28

Median H12 error 0.57 Å where the call is right, 9.27 Å where it is wrong.

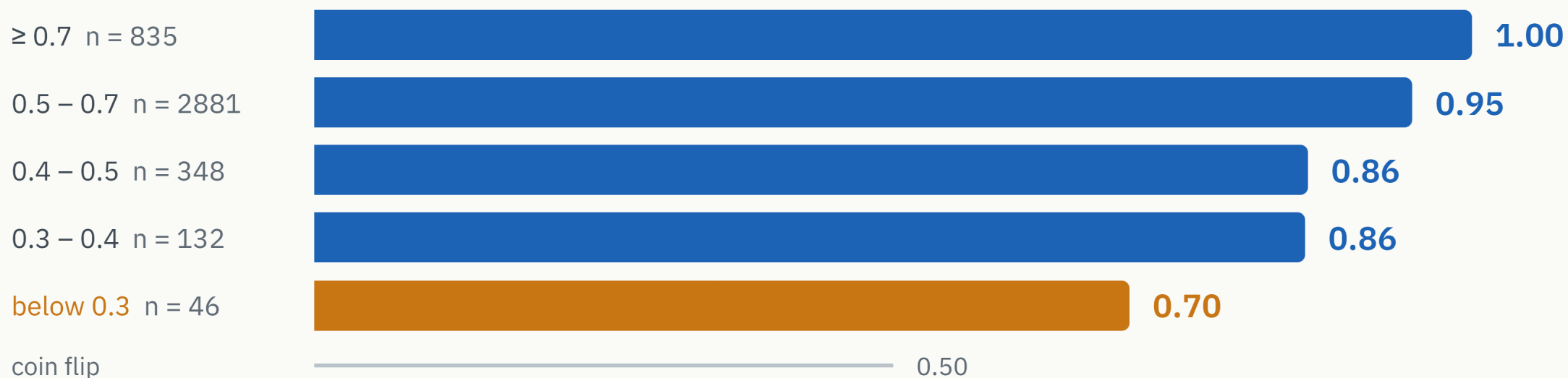
Oracle routing: 78 / 78

Generator error on the state call is exactly zero. Deeper sampling, better templates and pocket conditioning cannot buy a single point.

Read honestly: the 78/78 follows from the levers being absolute, so it localizes the error rather than discovering it — and these are structures the model has largely memorized, so the 0.57 Å is an optimistic geometry, not a claim about new targets.

The router reports how far it can see

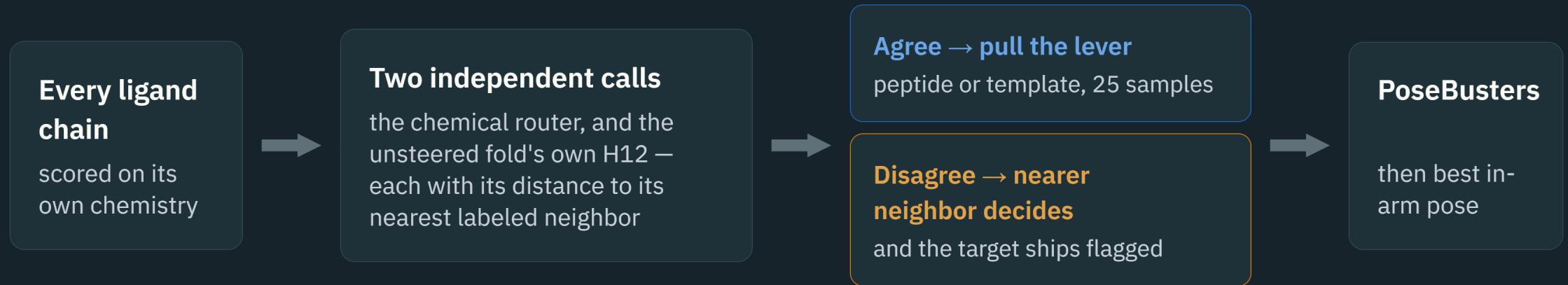
Ligand only, no GPU. Five chemical representations stacked, with the distance to the nearest labeled compound handed to the model as a feature — and reported with every call.



Below 0.3 the fingerprints die and plain physicochemical descriptors do not — 0.70 where every substructure model sits at chance. So the router switches representation on distance rather than trusting one model everywhere.

All nine holdout ligands sit at 0.50–0.80, in the top two bands. But the curve was fitted inside ChEMBL and does not transfer to crystal-ligand chemistry — there, a close neighbor did not protect against error.

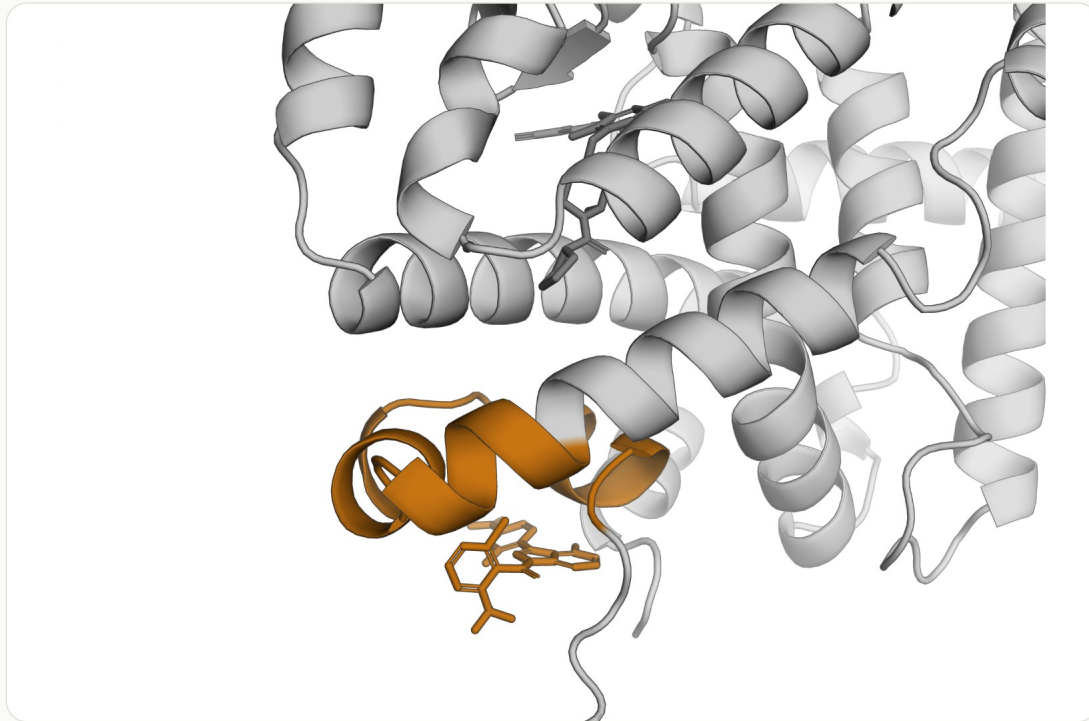
The method



Routing costs **no GPU**. Adding 91 structural features to the chemical model moves accuracy by 0.002 — the arms are downstream of the ligand the router already reads.

Co-ligands we were handed are scored, never read as a signal. A sterol in the query is a fact about the experiment, not about the compound.

The nine, and the two we argue about



Our own t09 prediction. Chain B sits orthosteric (grey); chain C is wedged against H12 (orange), 17 of its atoms within 4 Å. The dual-site architecture, in the target the router got wrong.

7 / 9

both calls agree — and so does the nearest crystal ligand

t05 → antagonist

Chemistry's neighbor sits at 0.800, the structure's at 0.403. Chemistry is inside its competence; the structure is not.

t09 → antagonist

Chain B is an exact match to an agonist entry's ligand and the model calls antagonist anyway. The cause is chain C — and a chain-B-only router cannot see it.

What we rejected, and what is next

Pocket conditioning

Inert on H12 across four runs, and fails PoseBusters 60–76 % of the time against 2.5–12 % unconstrained.

Deeper sampling

25 → 50 samples changed the call in 0 of 29 cells. Buy seeds, not samples.

Finetuning

The expensive option, and it does not touch routing — which is the entire measured gap.

Close the agonist gap

Every router error is an agonist chemotype with no ChEMBL counterpart. Adding the 77 deposited ligands targets exactly that, at no GPU cost.

Two questions for you

What is superposed before the H12 RMSD? And do the held-out targets include an orthosteric antagonist — none exists among the 78.

Generation is solved and costs nothing to point. The error that is left is a single binary call, and we can tell you which targets it is unsure about before you score us.